A purely analytic lower bound for the aggregate reserve at $K = 200$

Abstract

This note proves analytically one base-frequency sign in the aggregate-tail argument. Every definition is reproduced below. The proof uses one concave tangent bound, one Young inequality, elementary rational enclosures, and two exact Bernstein-basis identities. In fact, it proves the uniform, hand-checkable estimate

$$\mathcal{B}(\rho, 200) > 60 \qquad \left(\frac{7}{51} \le \rho \le \frac{39}{50}\right).$$

No decimal approximation or machine-generated sign enclosure is a premise.

1 The exact assertion

For $0 < \rho < 1$, put

$$G_1(s) = \frac{\sqrt{1-s^2} - s\arccos s}{\pi}, \qquad \eta_\rho = G_1\Big(\max\{\rho, \tfrac{1}{2}\}\Big), \qquad d_\rho = \frac{2\arcsin\rho}{\pi}.$$

At the fixed frequency $K = 200$, define

$$\mu = 200\rho, \qquad b = \frac{2\pi\rho}{1-\rho}, \qquad c = bK = \frac{400\pi\rho}{1-\rho}, \qquad \overline{R} = \mu + \frac{1}{2}, \qquad S = \sqrt{\overline{R}^2 + c},$$

and

$$\overline{\mathcal{I}} = \frac{1}{2}\left[\overline{R}S + c\,\operatorname{arsinh}\frac{\overline{R}}{\sqrt{c}} - \overline{R}^2\right]. \tag{1}$$

These are exactly the specializations at $K = 200$ of the aggregate-tail quantities in the main proof. The reserve to be estimated is

$$\begin{aligned}\mathcal{B}(\rho, 200) = (\mu - \tfrac{1}{2})(200\eta_\rho - 1) + \frac{d_\rho}{2}(\mu - \tfrac{3}{2})(\mu - \tfrac{1}{2}) \\ - (1 + d_\rho)\overline{\mathcal{I}} - \frac{8(\mu + \frac{1}{2})}{15\sqrt{\mu}}.\end{aligned} \tag{2}$$

Theorem 1. *For every*

$$\frac{7}{51} \le \rho \le \frac{39}{50},$$

one has

$$\mathcal{B}(\rho, 200) > 60 > 0.$$

2 A concave majorant for the integral term

Set

$$H(u) = \frac{1}{2}\left[u\sqrt{1+u^2} + \operatorname{arsinh} u - u^2\right] = \int_0^u (\sqrt{1+t^2} - t)\,dt.$$

Scaling (1) by $u = \overline{R}/\sqrt{c}$ gives the exact identity

$$\overline{\mathcal{I}} = cH(u). \tag{3}$$

Lemma 2 (Concave tangent)**.** *For every $u > 0$,*

$$H(u) < \frac{6}{25}u + \frac{12}{25}. \tag{4}$$

Proof. One has

$$H'(u) = \sqrt{1+u^2} - u, \qquad H''(u) = \frac{u}{\sqrt{1+u^2}} - 1 < 0.$$

Thus every tangent line is a global upper bound. Choose

$$u_0 = \frac{589}{300}, \qquad \sqrt{1+u_0^2} = \frac{661}{300}.$$

Then

$$H'(u_0) = \frac{6}{25}, \qquad \operatorname{arsinh} u_0 = \log\left(u_0 + \sqrt{1+u_0^2}\right) = \log\frac{25}{6},$$

and hence

$$H(u_0) - \frac{6}{25}u_0 = \frac{1}{2}\left(\log\frac{25}{6} - \frac{589}{1250}\right). \tag{5}$$

For completeness, the elementary series

$$\log 2 = 2\sum_{n=0}^{\infty} \frac{1}{(2n+1)3^{2n+1}}$$

gives

$$\log 2 < \frac{2}{3} + \frac{2}{3}\sum_{n=1}^{\infty} 3^{-(2n+1)} = \frac{25}{36}.$$

Also $\log(1+x) < x$ for $x > 0$. Therefore

$$\log\frac{25}{6} = 2\log 2 + \log\frac{25}{24} < \frac{25}{18} + \frac{1}{24} = \frac{103}{72} < \frac{1789}{1250};$$

the last difference is $29/45000$. Substitution in (5) makes the tangent intercept strictly smaller than $12/25$, proving (4) by concavity. $\square$

We shall rationalize the remaining square root using the elementary Young inequality

$$\sqrt{c} \le \frac{c}{2a} + \frac{a}{2} \qquad (a > 0), \tag{6}$$

which is equivalent to $(\sqrt{c} - a)^2 \ge 0$. The classical bound

$$\pi < \frac{22}{7} \tag{7}$$

follows, for example, from

$$0 < \int_0^1 \frac{x^4(1-x)^4}{1+x^2}\,dx = \frac{22}{7} - \pi.$$

Consequently

$$c < c_+(\rho) := \frac{8800}{7}\frac{\rho}{1-\rho}.$$

Combining Lemma 2 with (6) gives, for every $a > 0$,

$$\overline{\mathcal{I}} < Q_a(\rho) := \frac{3\overline{R}c_+(\rho)}{25a} + \frac{3a\overline{R}}{25} + \frac{12c_+(\rho)}{25}. \tag{8}$$

All quantities on the right are rational functions of ρ when a is rational.

3 The coefficient of d_ρ is positive

The replacement of d_ρ by an elementary lower bound requires the sign of its coefficient. We verify that sign before making any such replacement.

Lemma 3. *On* $[7/51, 39/50]$,

$$\frac{1}{2}(\mu - \tfrac{3}{2})(\mu - \tfrac{1}{2}) - \overline{\mathcal{I}} > 0. \tag{9}$$

Proof. Write

$$\Phi(\rho) = \overline{R}^2\frac{1-\rho}{\rho}.$$

Since

$$\frac{d}{d\rho}\log\Phi(\rho) = \frac{200\rho - 400\rho^2 - 1/2}{\rho(1-\rho)\overline{R}},$$

its two critical points are

$$q_\pm = \frac{10 \pm 7\sqrt{2}}{40}.$$

The elementary bounds $7/5 < \sqrt{2} < 3/2$ show

$$q_- < \frac{1}{200} < \frac{7}{51} < q_+ < \frac{41}{80} < \frac{39}{50}.$$

Thus Φ increases and then decreases on the interval in question, so its minimum is attained at one of the two endpoints. From (7),

$$u^2 = \frac{\overline{R}^2(1-\rho)}{400\pi\rho} > \frac{7\overline{R}^2(1-\rho)}{8800\rho}.$$

At the two endpoints the rational lower bounds on the right are

$$\frac{8128201}{2080800}, \qquad \frac{685783}{124800},$$

respectively, and

$$\frac{8128201}{2080800} - \frac{225}{64} = \frac{1625777}{4161600} > 0, \qquad \frac{685783}{124800} - \frac{225}{64} = \frac{247033}{124800} > 0.$$

Hence $u > 15/8$ throughout the interval.

By (3) and Lemma 2,

$$\frac{\overline{\mathcal{I}}}{\overline{R}^2} = \frac{H(u)}{u^2} < \frac{6}{25u} + \frac{12}{25u^2} \leq \frac{496}{1875}.$$

Moreover $\overline{R} \geq 2851/102 > 27$, and the function

$$\frac{(R-2)(R-1)}{2R^2} = \frac{1}{2} - \frac{3}{2R} + \frac{1}{R^2}$$

is increasing for $R > 4/3$. Therefore

$$\frac{(\mu - \frac{3}{2})(\mu - \frac{1}{2})}{2\overline{R}^2} \geq \frac{25 \cdot 26}{2 \cdot 27^2} = \frac{325}{729}.$$

Finally

$$\frac{325}{729} - \frac{496}{1875} = \frac{82597}{455625} > 0,$$

which proves (9). $\square$

4 Rational lower envelopes

We use only the consequences

$$\frac{1}{\pi} > \frac{7}{22}, \qquad \sqrt{2} > \frac{7}{5}, \qquad \sqrt{3} > \frac{5}{3}$$

of elementary positive-side squaring and (7).

On $7/51 \leq \rho \leq 1/2$, one has

$$\eta_\rho = G_1(1/2) = \frac{\sqrt{3}}{2\pi} - \frac{1}{6} > \frac{13}{132}, \qquad d_\rho = \frac{2\arcsin\rho}{\pi} > \frac{7}{11}\rho.$$

Also $1400/51 \leq \mu \leq 100$, and $(26/5)^2 < 1400/51$, so

$$\frac{8(\mu + 1/2)}{15\sqrt{\mu}} = \frac{8}{15}\left(\sqrt{\mu} + \frac{1}{2\sqrt{\mu}}\right) < \frac{8}{15}\left(10 + \frac{5}{52}\right) = \frac{70}{13}. \tag{10}$$

Define

$$\eta_< = \frac{13}{132}, \qquad d_< = \frac{7}{11}\rho, \quad E_< = \frac{70}{13},$$

$$L_<(\rho) = (\mu - \tfrac{1}{2})(200\eta_< - 1) + \frac{d_<}{2}(\mu - \tfrac{3}{2})(\mu - \tfrac{1}{2}) - (1 + d_<)Q_{18}(\rho) - E_<. \tag{11}$$

On $1/2 \leq \rho \leq 39/50$, the integral representation

$$\eta_\rho = \frac{1}{\pi}\int_\rho^1 \arccos t \, dt$$

and $\arccos t \geq \sqrt{2(1-t)}$ give

$$\eta_\rho \geq \frac{2\sqrt{2}}{3\pi}(1-\rho)^{3/2}.$$

Here $1-\rho \geq 11/50 > (23/50)^2$, and consequently

$$\eta_\rho > \frac{1127}{8250}(1-\rho) =: \eta_>. \tag{12}$$

The Maclaurin series for $\arcsin \rho$ has positive coefficients, so

$$d_\rho > \frac{7}{11}P_7(\rho) =: d_>, \qquad P_7(\rho) = \rho + \frac{\rho^3}{6} + \frac{3\rho^5}{40} + \frac{5\rho^7}{112}. \tag{13}$$

Finally $100 \leq \mu \leq 156 < 169$, whence

$$\frac{8}{15}\left(\sqrt{\mu} + \frac{1}{2\sqrt{\mu}}\right) < \frac{8}{15}\left(13 + \frac{1}{20}\right) = \frac{174}{25} < 7. \tag{14}$$

Define

$$\begin{aligned} E_> &= 7, \\ L_>(\rho) &= (\mu - \tfrac{1}{2})(200\eta_> - 1) + \frac{d_>}{2}(\mu - \tfrac{3}{2})(\mu - \tfrac{1}{2}) - (1 + d_>)Q_{67}(\rho) - E_>. \end{aligned} \tag{15}$$

Lemma 3, the lower bounds for d_ρ and η_ρ, the upper bounds for the last term, and (8) now give the logically ordered implications

$$\mathcal{B}(\rho, 200) > L_<(\rho) \quad \left(\frac{7}{51} \leq \rho \leq \frac{1}{2}\right), \qquad \mathcal{B}(\rho, 200) > L_>(\rho) \quad \left(\frac{1}{2} \leq \rho \leq \frac{39}{50}\right). \tag{16}$$

The overlap at $\rho = 1/2$ is harmless.

5 Exact Bernstein closure

For $0 \leq x \leq 1$, write

$$B_{k,n}(x) = \binom{n}{k} x^k (1-x)^{n-k}.$$

These functions are nonnegative and sum to one. The conversion from a power polynomial $\sum_{j=0}^n q_j x^j$ to Bernstein coefficients is the finite rational identity

$$\sum_{j=0}^{n} q_j x^j = \sum_{k=0}^{n} \left(\sum_{j=0}^{k} \frac{\binom{k}{j}}{\binom{n}{j}} q_j \right) B_{k,n}(x). \tag{17}$$

Thus every entry below can be checked by additions and cross multiplications of integers.

For the lower interval set

$$x_< = \frac{\rho - 7/51}{1/2 - 7/51} = \frac{102\rho - 14}{37}.$$

Direct substitution of (8) and (11), followed by (17), gives the exact identity

$$(1-\rho)L_<(\rho) = \frac{1}{D_<}\sum_{k=0}^{4} N_{<,k} B_{k,4}(x_<), \qquad D_< = 10835145921600, \tag{18}$$

where

k	$N_{<,k}$
0	2977383044192480
1	4628920786869499
2	5747338995812809
3	6988840660610010
4	6621947656848702

The smallest entry is $N_{<,0}$, and

$$N_{<,0} - 60D_< = 2327274288896480 > 0.$$

It follows from (18) that

$$(1-\rho)L_<(\rho) > 60. \tag{19}$$

For the upper interval set

$$x_> = \frac{\rho - 1/2}{39/50 - 1/2} = \frac{25}{7}\left(\rho - \frac{1}{2}\right).$$

The corresponding exact identity obtained from (8), (15), and (17) is

$$(1-\rho)L_>(\rho) = \frac{1}{D_>}\sum_{k=0}^{10} N_{>,k} B_{k,10}(x_>), \qquad D_> = 248737500000000000000, \tag{20}$$

where

k	$N_{>,k}$
0	873011369240936279296875
1	881095006796855712890625
2	884923416376472412109375
3	881704924277462068359375
4	867541422038412498046875
5	837327296894843418528125
6	784608885866591053548975
7	701387295930232136634075
8	577837649792730661053099
9	401902155200474706698937
10	158688669397963044788415

The smallest entry is $N_{>,10}$, and

$$N_{>,10} - 60D_> = 9446169397963044788415 > 0.$$

Therefore

$$(1-\rho)L_>(\rho) > 60. \tag{21}$$

Since $0 < 1-\rho < 1$, both (19) and (21) imply that the corresponding L-function is strictly larger than 60. Combining this conclusion with (16) proves Theorem 1.

Verification boundary. The two Bernstein identities are finite polynomial identities over $\mathbb{Q}$. Formula (17) is a direct hand-checking recipe for every printed integer. No interval evaluation, floating-point comparison, or unprinted case is used in the proof.